

Readable but Not Steerable: Evidence Sufficiency and Uptake in the Residual Stream of Retrieval-Augmented LLMs

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Abstract

When retrieved evidence becomes sufficient to answer a question, does a language model’s internal state predict, before it generates a token, whether it will use that evidence? We run paired forward passes over *evidence trajectories* that add distractors, individual reasoning hops, the decisive hop, redundant, falsified, partial or paraphrased support, or the answer string in an irrelevant passage, and read the residual stream at the answer-start position. Linear probes on the paired update $\Delta h = h(C_k) - h(C_{k-1})$ recover *sufficiency* at 0.95 AUROC on document-, answer- and relation-disjoint splits in three model families, but a fine-tuned text classifier recovers it as well. A nearly orthogonal direction predicts *uptake*, whether decisive evidence will be integrated rather than ignored, at 0.78 AUROC, where text classifiers stay at 0.52–0.63: uptake is an internal signal. Causal tests separate readout from mechanism. Probe directions are inert under steering, projection-out and component patching; the full anchor residual is causally sufficient and its effect lies in the orthogonal complement; the only causally active direction, the sufficient-minus-insufficient mean difference, is an abstention gate that raises answering on distractors and false evidence as much as on sufficient evidence. These results replicate on HotpotQA, on RAGBench’s naturally retrieved documents across five domains, on BM25- and E5-retrieved Wikipedia evidence, and on a synthetic benchmark, although Wikipedia-trained directions transfer only partially across domains. High-AUROC, transferable probes find the label, not the lever.

1 Introduction

Retrieval-augmented generation supplies a language model with passages and asks it to answer from them. Two internal questions matter for such systems: has the model registered that the supplied evidence is now *sufficient*, and has it *integrated* the decisive passage into its answer computation? A reliable pre-generation readout of either would let a system decide whether to retrieve again, detect ignored evidence, or abstain; a causal handle on either would let a system correct the model’s behaviour directly. Prior work has shown that residual-stream activations carry signals about context-versus-parametric conflict and context reliance, but these are typically identified on single contexts, and it is rarely tested whether the directions that read such states also control them.

We take a different unit of analysis: the *paired update*. For each multi-hop question we construct nested evidence contexts $C_0 \subset C_1 \subset \dots$ and run a fresh, stateless forward pass on each, reading the residual stream at the fixed answer-start position. The difference $\Delta h_k = h(C_k) - h(C_{k-1})$ isolates what one added paragraph did to the model’s state. Because every decisive increment (the paragraph completing the evidence) is matched with controls that add a length-matched distractor, a duplicate of already-present support, a falsified version of the decisive paragraph, or the answer string embedded in an irrelevant passage from the same penultimate state, a probe trained on these updates cannot succeed by detecting length, answer-string presence or passage relevance alone. Each increment is also labelled with the model’s own behavioural transition (wrong→correct, wrong→wrong, correct→wrong, and so on), giving separate supervision for sufficiency, uptake, correction and stability rather than one conflated “answer-update” target.

Our contributions are as follows.

- **A matched-increment protocol** for reading evidence-sufficiency and evidence-uptake states from single forward passes, with four-way disjoint evaluation (question, document, answer entity, relation type) and a conversational-revision ablation (Sections 2 and 3).

- **Readout results.** Sufficiency is linearly decodable at 0.95 AUROC in Qwen2.5-7B, Llama-3.1-8B and Mistral-7B, survives residualising twelve nuisance covariates, orders increment kinds as predicted, and transfers without retraining to a conversational protocol, to 2WikiMultiHopQA and to 3–4-hop questions. Uptake is decodable at 0.78 along a nearly orthogonal direction, and unlike sufficiency it is not recoverable from the text (0.56 for a fine-tuned DeBERTa) (Section 4).
- **Causal dissociation.** Probe directions are inert under steering, projection-out and component patching in all three models; the full anchor residual is causally sufficient, its effect lives in the orthogonal complement, and the one causally active direction (the mean difference) is an abstention gate, not an accommodation mechanism (Section 5).
- **External validity.** Everything replicates on HotpotQA, on RAGBench’s naturally retrieved documents across five domains and single-hop questions, and on a synthetic benchmark with paraphrase, partial and contradictory evidence; Wikipedia-trained directions transfer only partially across domains (Section 6).

The picture that emerges is that evidence sufficiency and uptake are cleanly *readable* before generation, that the readout of uptake is not recoverable from text, and that neither readout is a *lever*: the linear probe finds a correlate of the state, while the mechanism that turns evidence into an answer is high-dimensional and content-bearing.

2 Experimental setup

Data. MuSiQue [2] provides 2–4-hop questions with gold supporting paragraphs, per-hop decompositions with sub-answers and relation types, and distractors; our main runs use its two-hop training questions, a second run 3–4-hop questions, and 2WikiMultiHopQA [1] is the transfer set. For each model we screen a pool of 4,000 questions closed-book and keep 2,000 that the model answers incorrectly without evidence (closed-book accuracy is 1.4% for Qwen, 4.0% for Llama, 1.3% for 3–4-hop questions), so every increment concerns evidence the model actually needs. We keep the whole trajectory for every question, including stages after the answer first becomes correct, to avoid conditioning the data on the model’s behaviour.

Models. Qwen2.5-7B-Instruct (28 layers, $d = 3584$), Llama-3.1-8B-Instruct (32 layers, $d = 4096$) and Mistral-7B-Instruct-v0.3 (32 layers, $d = 4096$), all in bfloat16 with greedy decoding of at most 16 tokens. Prompts use each model’s chat template.

Prompt and anchor. Every stateless stage uses the same template: an instruction (“Use only the supplied evidence to answer the question. If the evidence is insufficient, output INSUFFICIENT.”), the evidence block, the question and an answer cue. We read the residual stream after every decoder block at the last prompt token, the position that emits the first answer token, so all readouts are pre-generation. An *evidence-augmented* regime (“use the evidence together with your own knowledge”) is run in parallel and gives the same results (Appendix E); we report the evidence-only regime unless stated.

Evidence conditions. For a question with gold hop paragraphs $g_1, \dots, g_n$ we build (Appendix A): no evidence; matched distractors only; prefix chains in the canonical, reverse and (for $n \geq 3$) random hop orders, the last prefix being *minimally sufficient*; leave-one-out contexts; the sufficient context plus a distractor, a duplicate support or a conflicting falsified paragraph; and the full 20-paragraph context. New paragraphs are inserted at a random position, so increments are textually minimal while the decisive paragraph’s position is randomised. Falsified paragraphs swap the hop’s sub-answer for another question’s sub-answer of the same relation type; answer-string controls append a “not to be confused with {answer}” hatnote to a distractor.

Increments and labels. From every penultimate state S (all hops but one) we form the decisive increment $S \rightarrow S + g_{\text{last}}$ and four matched controls from the same S : distractor, redundant, false and answer-string. With first-hop, post-sufficiency and full-context increments this yields 17 conditions and 17 increments per two-hop question (35,008 conditions per model). Each increment carries $y^{\text{suff}} = \mathbf{1}[C_k \text{ sufficient}, C_{k-1} \text{ not}]$

and the model’s behavioural transition, defining *uptake* (decisive increments: answer becomes correct or not), *correction* (wrong or abstaining $\rightarrow$ correct or not), *stability* (correct $\rightarrow$ correct or not) and *revision* (answer changed). Correctness is exact match on any alias or containment within a short prediction.

Behaviour and splits. All three models abstain on 94–97% of distractor-only contexts and answer 64–75% of minimally sufficient ones with 36–47% correct; falsified final hops halve abstention (Table 5, Appendix C). Probes are trained on 70% of questions and evaluated on the rest under four groupings never shared between train and test: question id, gold document (connected components of questions sharing a supporting page), answer entity, and relation-type tuple. We report the document split by default and all four in Table 6 (Appendix D).

3 Method

Paired updates and probes. Let $h^{(l)}(C) \in \mathbb{R}^d$ be the residual stream after block l at the anchor for context C . For increment $C_{k-1} \rightarrow C_k$ we form $\Delta h^{(l)} = h^{(l)}(C_k) - h^{(l)}(C_{k-1})$ and, for each task and layer, fit an ℓ_2 -regularised logistic regression on standardised features with class balancing, selecting the layer by grouped three-fold cross-validation on training questions only. We also report the *difference-of-means* direction $v_{\text{dm}} \propto \mathbb{E}[\Delta h \mid y=1] - \mathbb{E}[\Delta h \mid y=0]$ and, for the raw state $h(C)$, *state probes* for sufficiency and correctness. Directions are unit vectors in raw activation space.

Nuisance analysis. A paired difference also encodes added tokens, context length, answer-string presence, passage relevance and the change in the output distribution. We build a nuisance vector z per increment (token counts, gold-answer log-probability change, next-token entropy and top-probability change, answer-string presence, TF-IDF relevance, insertion position, and a 16-dimensional TF-IDF-SVD embedding of the added text; Appendix B) and report (i) a nuisance-only classifier, (ii) the probe on residuals of a ridge regression of Δh on z , and (iii) the AUROC gain of adding out-of-fold probe scores to the nuisance model.

Text baselines. Text-only classifiers on the same increments and splits: TF-IDF logistic regression and a fine-tuned DeBERTa-v3-base cross-encoder, on “question [SEP] added paragraph” alone and with the previous context appended.

Transfer. Learned directions are applied without retraining to (i) the *conversational revision* protocol, in which the model sees its own previous answer A_k and only the new evidence and is told to change its answer only if the total evidence warrants it, pairing $h_{\text{conv}}(C_k \rightarrow C_{k+1}) - h(C_k)$; (ii) 2WikiMultiHopQA; (iii) 3–4-hop MuSiQue; and (iv) the datasets of Section 6. Model-family transfer is replication of the procedure, since hidden sizes differ.

Causal tests. Interventions act on held-out questions of the document split. *Steering* adds $\alpha u v$ from the anchor onward at layer l (or all positions), where u is the separation between the sufficient and insufficient state means along v on training data, so $\alpha = 1$ moves an activation by one sufficient-minus-insufficient gap; a random unit direction with the same ℓ_2 magnitude is the control. *Projection-out* sets the component along v to its insufficient-state mean at all layers from l onward. *Patching* runs the insufficient prompt C_{n-1} with its anchor residual replaced, at every layer from l onward, by h_{suff} (full), $h_{\text{ins}} + \text{proj}_v(\Delta)$ (parallel), $h_{\text{ins}} + \Delta - \text{proj}_v(\Delta)$ (orthogonal), the projection of Δ onto a random direction, onto the top- k principal components of training deltas, or onto the gold answer’s first-token unembedding direction. Outcomes are the rate of non-abstention (“answered”) and of correct answers on 150 held-out questions per set; for false-evidence contexts we also report the rate of producing the injected false answer.

4 Readout results

Sufficiency is linearly readable and robust. Table 1 shows that the decisive increment is separated from all non-decisive increments at 0.948–0.952 AUROC in the three models, with the best layer at 50–60%

Task	Qwen2.5-7B			Llama-3.1-8B			Mistral-7B		
	AUROC [95% CI]	nuis.	gain	AUROC [95% CI]	nuis.	gain	AUROC [95% CI]	nuis.	gain
Sufficiency	0.948 [0.94,0.95]	0.87	0.11	0.952 [0.95,0.96]	0.86	0.12	0.949 [0.94,0.95]	0.87	0.11
Sufficiency (matched)	0.911 [0.90,0.92]	0.90	0.07	0.918 [0.91,0.93]	0.89	0.08	0.910 [0.90,0.92]	0.90	0.07
Uptake	0.775 [0.74,0.80]	0.66	0.14	0.783 [0.75,0.81]	0.58	0.21	0.771 [0.74,0.80]	0.59	0.18
Correction	0.730 [0.70,0.76]	0.81	0.01	0.817 [0.79,0.84]	0.83	0.06	0.792 [0.77,0.81]	0.85	0.03
Stability	0.750 [0.72,0.78]	0.92	-0.01	0.869 [0.85,0.89]	0.90	0.02	0.800 [0.77,0.83]	0.90	0.01
State sufficiency	0.992 [0.99,0.99]	—	—	0.994 [0.99,0.99]	—	—	0.996 [1.00,1.00]	—	—

Table 1: Held-out AUROC of the best-layer probe on paired activation updates (document-disjoint split; 95% question-level bootstrap intervals), with the AUROC of a classifier on nuisance covariates alone and the gain from adding the probe score to that classifier. State-sufficiency probes act on the raw activation.

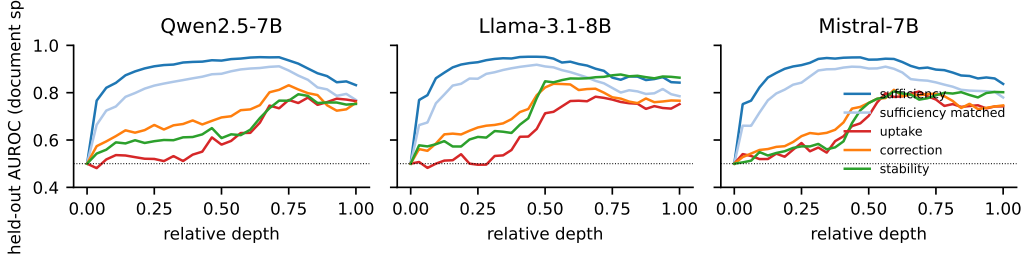


Figure 1: Held-out AUROC by relative depth for the update tasks (document split). Sufficiency peaks around mid-depth in every model; uptake and correction emerge later.

depth (Fig. 1). Against the matched controls only (distractor, redundant, false, answer-string from the same penultimate state) the probe still reaches 0.91–0.92. The direction survives residualising the nuisance covariates (0.94–0.96) and adds +0.11 AUROC to a nuisance-only classifier. Table 6 shows the result is insensitive to the grouping: question-, document-, answer-entity- and relation-disjoint splits all give 0.94–0.96, and the state probe on the raw activation reaches 0.99. Projections of held-out increments on the sufficiency direction order the increment kinds exactly as hypothesised (Fig. 2): decisive > false > distractor > redundant > answer-string control > non-decisive gold, with post-sufficiency increments and the first hop far below. The false-evidence increment scoring second indicates that the direction encodes “the final slot has been filled” rather than truth, which is what the model can know before generation.

Uptake is a separate, later signal. Whether a decisive increment is actually integrated (answer becomes correct) versus behaviourally ignored is decodable at 0.77–0.78 AUROC (document split; 0.85 question split), at layers near the end of the network, and this is the task where the activation adds most beyond nuisance covariates (+0.14 to +0.21). The uptake direction is nearly orthogonal to the sufficiency direction (cosine 0.02–0.03) and the sufficiency direction reaches only 0.54–0.60 on the uptake task, while uptake and wrong-to-correct *correction* share a direction (cosine 0.30–0.34; cross-task AUROC 0.82; Fig. 5, Appendix F). Correction and stability, by contrast, are largely explained by the change in the output distribution: nuisance-only classifiers reach 0.81–0.85 and 0.90–0.92, and the probe adds at most 0.06.

Transfer. Applying the 2-hop stateless directions without retraining (Table 7, Appendix F): To the conversational protocol they transfer at 0.79 (sufficiency), 0.78 (uptake), 0.92 (correction) and 0.82 (stability), supporting the view that the transition is intrinsic to evidence integration rather than an artefact of being asked to reconsider. To 2WikiMultiHopQA the sufficiency direction transfers at 0.96 (0.93 matched), correction at 0.85 and uptake at 0.79; probes trained within 2WikiMultiHopQA reach 0.99, 0.89 and 0.91. To 3–4-hop questions sufficiency transfers at 0.82–0.85; within that run the leave-one-out completions score as high as chain completions (Appendix H), and sufficiency probes reach 0.93 on every split, but uptake falls to 0.61 on the document and answer splits (from 0.80 on the question and relation splits).

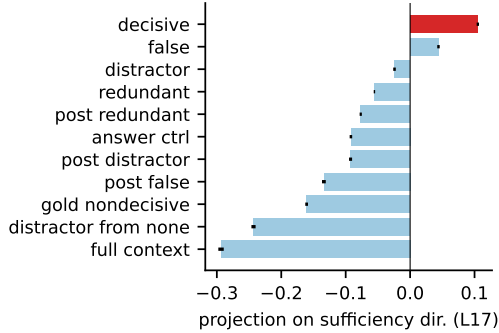


Figure 2: Mean projection of held-out increments on the sufficiency direction by kind (Qwen2.5-7B, document split; standard errors). Red: decisive.

Model	Budget	Task	TF-IDF	DeBERTa (q+added)	DeBERTa (+context)	Probe
Qwen2.5-7B	reduced budget	sufficiency	0.623	0.723	0.900	0.948
Qwen2.5-7B	reduced budget	sufficiency matched	0.623	0.758	0.823	0.911
Qwen2.5-7B	reduced budget	uptake	0.574	0.569	0.557	0.775
Qwen2.5-7B	full budget	sufficiency	0.611	0.723	0.972	0.948
Qwen2.5-7B	full budget	uptake	0.574	0.625	0.630	0.775
Llama-3.1-8B	reduced budget	sufficiency	0.625	0.715	0.888	0.952
Llama-3.1-8B	reduced budget	uptake	0.463	0.466	0.517	0.783
Mistral-7B	reduced budget	sufficiency	0.625	0.730	0.920	0.949
Mistral-7B	reduced budget	uptake	0.536	0.573	0.604	0.771

Table 2: Text-only classifiers on the same increments and splits versus the activation probe (document split).

4.1 Is the internal state needed? Text-only baselines

Table 2 answers the obvious objection. For sufficiency, TF-IDF reaches 0.62 and a DeBERTa that sees only the question and the added paragraph 0.72–0.73, but a DeBERTa that also sees the previous context reaches 0.89–0.92 at a reduced training budget (one epoch, 256 tokens, 8k increments) and 0.97 at full budget (two epochs, 512 tokens, all increments), exceeding the probe’s 0.95 on the same split. Sufficiency is therefore largely recoverable from the text: the activation probe is an accurate readout of a property the input already determines. Uptake is the opposite case. On the same increments no text model exceeds 0.63 on whether the model will integrate the decisive paragraph (Qwen 0.56 reduced and 0.63 full budget, Llama 0.52, Mistral 0.60 with context), while the activation probes reach 0.77–0.78: whether evidence is taken up depends on the model’s internal state, not on the evidence alone.

5 Causal results: readout versus mechanism

Probe directions are inert. Additive steering along the logistic sufficiency or uptake direction at the best layer changes no aggregate rate at any dose from 0.7 to 22 ℓ_2 units (35% of the residual norm, Appendix K), and a matched random direction is equally inert, so the null is not a matter of dose. Patching only the component of the sufficient-minus-insufficient update that lies along the probe direction, at every layer from the best one onward, leaves the insufficient prompt’s behaviour unchanged in all three models (Table 3, “ $\parallel v$ ”); projecting the logistic direction out of sufficient prompts changes nothing either (v_{lr} column). The same holds for the uptake direction.

The anchor residual is causally sufficient, in the orthogonal complement. Replacing the anchor residual of the insufficient prompt with the sufficient run’s, at all layers from the best layer onward, takes

Model	Patching the insufficient prompt (correct/answered)							Projection-out on sufficient (answered)			
	none	full	$\parallel v$	$\perp v$	rand-256	PC1	PC256	unemb.	v_{dm}	random	v_{lr}
Qwen2.5-7B	0.04/0.13	0.26/0.71	0.04/0.13	0.25/0.71	0.04/0.13	0.09/0.29	0.18/0.59	0.09/0.19	0.01	0.72	0.72
Llama-3.1-8B	0.02/0.11	0.24/0.74	0.02/0.12	0.24/0.73	0.02/0.12	0.02/0.19	0.09/0.41	0.19/0.62	0.00	0.73	0.75
Mistral-7B	0.07/0.17	0.09/0.60	0.07/0.17	0.09/0.60	0.07/0.17	0.08/0.61	0.08/0.56	0.07/0.19	0.01	0.63	0.63
Qwen 3-4 hop	0.05/0.20	0.25/0.62	0.06/0.19	0.25/0.62	0.06/0.20	0.07/0.23	0.20/0.51	0.07/0.21	0.03	0.63	0.63

Table 3: Causal tests, 150 held-out questions, all layers from the best sufficiency layer onward. Left: insufficient prompt with its anchor residual patched (correct / answered). Right: sufficient prompt with the component along a direction set to its insufficient-state value (answered). v_{dm} : mean-difference direction; v_{lr} : logistic-probe direction.

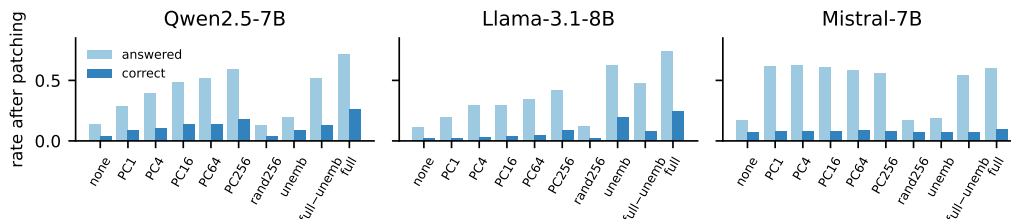


Figure 3: Rank of the causal subspace: the insufficient prompt patched with the sufficient update projected on the top- k principal components of training updates, on a random 256-dimensional subspace, on the gold answer’s unembedding direction (“unemb”), or on everything but that direction.

Qwen from 0.04/0.13 (correct/answered) to 0.26/0.71 and Llama from 0.02/0.11 to 0.24/0.74, roughly half the correctness the model achieves when the paragraph is actually present. Removing the probe component from that patch (“ $\perp v$ ”) leaves the effect intact (0.25/0.71, 0.24/0.73). The single anchor position therefore carries enough to make the model answer, and often answer correctly, but none of that content is along the direction the probe reads. Mistral shows the same dissociation with a weaker full-patch effect (0.09/0.60), as does the 3-4-hop run (0.25/0.62).

The causal content is high-rank and model-specific. Figure 3 patches the update projected on the top- k principal components of training updates. Answering rises gradually with rank (Qwen: 0.29 at $k=1$, 0.49 at $k=16$, 0.59 at $k=256$, 0.71 full) while a random 256-dimensional subspace does nothing; correctness needs even more of the update. The first component carries much of the answer-versus-abstain switch, consistent with the projection-out result below. What carries correctness differs by model: on Llama the gold answer’s unembedding direction alone restores 0.19/0.62, most of the full effect, whereas on Qwen and Mistral it restores little (0.09/0.19, 0.07/0.19) although removing it from the full patch halves Qwen’s correctness (0.26 $\rightarrow$ 0.13). Mistral’s and the multi-hop run’s effects are captured by the top principal components instead (0.56 and 0.51 answered at $k=256$).

The mean-difference direction is a causal abstention gate. The one direction with a causal effect is v_{dm} , the mean difference between sufficient and insufficient states, which the discriminative probe does not align with. Setting its component to the insufficient-state value at all layers from the best one onward makes all three models abstain on essentially every sufficient prompt (answered 0.01, 0.00, 0.01 versus 0.72, 0.73, 0.63 for a random direction under identical treatment), and equally on false-evidence prompts. Low-dose additive steering along v_{dm} over four layers (Fig. 4; full numbers in Table 8, Appendix L) gives a graded, coherent, monotone dose-response in both directions with a flat random control: at $\alpha = -1$ Qwen answers 34% of sufficient prompts (from 71%) with unchanged precision among the answers it gives. The positive direction, however, is indiscriminate. At $\alpha = +1$ answering rises on distractor-only prompts (Qwen 0.03 $\rightarrow$ 0.12, Llama 0.03 $\rightarrow$ 0.29, Mistral 0.03 $\rightarrow$ 0.28), on insufficient prompts (0.13 $\rightarrow$ 0.26, 0.11 $\rightarrow$ 0.51, 0.17 $\rightarrow$ 0.47), on answer-string controls, and false-evidence following roughly doubles on Llama (0.19 $\rightarrow$ 0.39), while correct answers on sufficient prompts gain 3 points on Qwen and 18 on Llama. By the criterion we

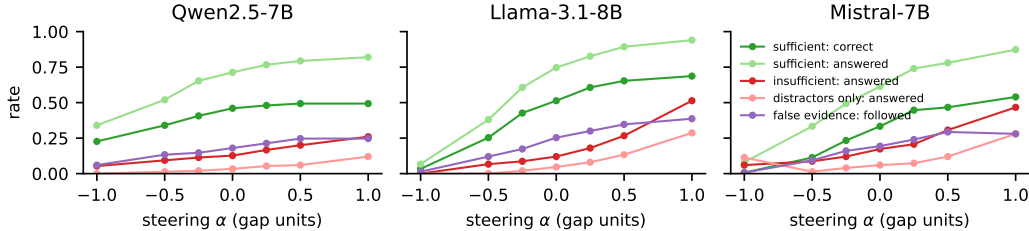


Figure 4: Dose-response of difference-of-means steering (four layers from the best sufficiency layer, anchor onward; α in units of the sufficient-minus-insufficient gap). Negative α induces abstention everywhere; positive α induces answering everywhere, including on distractors and false evidence.

set in advance, a direction that makes the model accept any passage is a context-compliance or abstention direction, not an accommodation direction. Applied across all positions and all layers at $|\alpha| \geq 2$ the same recipe destroys the output for probe and random directions alike (Appendix L).

Used to stop retrieval, the pre-generation sufficiency readouts match “stop when the model does not say INSUFFICIENT” on MuSiQue and beat it on HotpotQA (0.73 vs 0.65); uptake, correction and stability are poor stopping rules, but the uptake (or correction) direction is the best learned verifier of the answer once retrieval stops (Appendix N).

6 External validity: other benchmarks, natural retrieval, domain shift

We repeat the pipeline unchanged on three further sources (Table 4): **HotpotQA** (distractor setting; 2,000 questions, closed-book-correct ones kept); **RAGBench**, 3,000 mostly single-hop questions from twelve sub-datasets across biomedical, finance, technical, customer-support and general domains, whose evidence units are the *naturally retrieved* documents of the original RAG pipelines (gold: documents with annotated relevant sentences; distractors: the other retrieved documents); and a **synthetic** benchmark of 2,400 questions over fictional entities (closed-book accuracy zero) with one to three hops, five template families per relation, and paraphrase, partial and contradictory variants of every gold fact.

Readouts. Sufficiency is decodable on every source (0.98 HotpotQA, 0.90 RAGBench, 0.97 synthetic) and against matched controls (0.96, 0.94, 0.95), which on the synthetic data include paraphrastic redundancy, partial evidence and contradictions; the projection ordering there is decisive (chain, leave-one-out or paraphrased) $>$ contradictory $>$ partial $\approx$ redundant $>$ distractor $>$ answer-string control. Uptake is decodable on all three (0.73, 0.69, 0.93; RAGBench labels from an LLM judge against its long-form references), and the per-group breakdown (Table 11, Appendix O) shows it is not a multi-hop artefact: on RAGBench single-hop questions it reaches 0.70 ($n=444$), on HotpotQA comparison questions, whose two facts are independent lookups, 0.79 against 0.68 for bridge questions, and on single-hop synthetic questions 0.99. It is above chance in every RAGBench domain (0.56–0.78) and, on HotpotQA, in both closed-book-correct (0.80) and closed-book-incorrect (0.72) questions, while sufficiency is decoded equally in both groups (0.984 vs 0.978): the readout is not a “need for evidence” signal.

Transfer across construction and domain. The MuSiQue directions transfer to HotpotQA at 0.95 (sufficiency) but only partially to RAGBench (0.71; matched 0.78; uptake 0.55) and to the synthetic data (0.74; uptake 0.72; correction 0.80). Within RAGBench, holding out whole sub-datasets drops sufficiency from 0.90 to 0.72 (matched 0.94 $\rightarrow$ 0.87, uptake 0.69 $\rightarrow$ 0.60), and holding out relation chains on the synthetic data costs two to five points. The state is therefore present under natural retrieval and in every domain, but its direction is partly dataset-specific; a Wikipedia-trained detector should not be deployed unadapted on biomedical or technical RAG.

Dataset	within-dataset probes (AUROC)				MuSiQue dir.		patched insufficient (answered)				proj.-out (answered)	
	suff.	matched	uptake	corr.	suff.	uptake	none	full	$\parallel v$	$\perp v$	v_{dm}	random
HotpotQA	0.979	0.961	0.729	0.846	0.952	0.649	0.41	0.92	0.41	0.92	0.17	0.92
RAGBench	0.898	0.944	0.685	0.742	0.706	0.550	0.15	0.72	0.15	0.72	0.10	0.73
RAGBench (held-out sub-datasets)	0.717	0.872	0.596	0.719	0.706	0.550	—	—	—	—	—	—
Synthetic	0.965	0.948	0.930	0.925	0.739	0.722	0.03	0.89	0.03	0.89	0.00	0.99
Synthetic (held-out relation chains)	0.945	0.925	0.883	0.897	0.739	0.722	—	—	—	—	—	—

Table 4: External validity (Qwen2.5-7B): within-dataset probes on the row’s split, MuSiQue 2-hop directions applied without retraining, and the patching and projection-out tests of Section 5 (answered rate, 150 held-out questions).

Causal dissociation. The dissociation replicates on all three sources: the full anchor patch makes the insufficient prompt answer ($0.41 \rightarrow 0.92$, $0.15 \rightarrow 0.72$, $0.03 \rightarrow 0.89$), the probe-direction component does nothing, the orthogonal component carries the effect, and projecting out v_{dm} silences answering (0.17, 0.10, 0.00) while a random direction does not. On synthetic data the patched model answers 89% of the time but is never correct: the anchor carries the decision to answer, while a fictional answer can only be read from evidence tokens the insufficient prompt lacks.

Evidence State Transition Benchmark: retrieval source and linguistic form. We rebuild MuSiQue trajectories with evidence from the 21M-passage Wikipedia corpus of FlashRAG, retrieved by BM25 or E5 with the question plus one query per hop: either the gold paragraphs are kept and all distractors are retrieved, or *all* evidence is retrieved (402 questions for E5, 219 for BM25, where retrieval found every hop). A further run adds LLM-paraphrased and value-withheld (partial) versions of each gold paragraph and independently authored paragraphs on the same entity from HotpotQA or 2WikiMultiHopQA. Nothing changes qualitatively (Table 15; construction, measurements and full results in Appendix P): sufficiency stays at 0.91–0.95, the MuSiQue directions transfer at 0.91–0.94 (sufficiency) and 0.76–0.82 (uptake) on questions the source probes never saw, and the causal dissociation and abstention gate reproduce in every cell. Paraphrased and independently authored decisive paragraphs project like the originals; partial and contradictory ones are the hardest controls (0.81, 0.76; Table 16).

7 Discussion

Two signals, one lever, and none of them is accommodation. The results separate three things that “an internal state transition” conflates. *Sufficiency* is a state of the input that the model represents cleanly and a text model can also compute; its causal counterpart is an abstention gate. *Uptake* is a state of the model: whether the same decisive paragraph will be integrated is predictable from activations and not from text, at 0.78 AUROC across three families and on every external source. Neither readout direction is a lever; what turns an insufficient state into an answer is high-rank, orthogonal to both, and on some models concentrated on the answer’s unembedding direction. The picture is a low-dimensional gate (answer or abstain) plus high-dimensional answer content, not a shared accommodation direction.

Why probes do not find the lever, and what to use. A logistic probe maximises separability along directions of low within-class variance; the mean-difference direction has a large component along a high-variance “answering” axis that the discriminative weights down-weight, and the projection-out and rank experiments show the causal content is exactly there. A high-AUROC, transferable probe direction can thus be causally irrelevant, which argues for validating any proposed steering direction with the projection-out, rank and matched-random controls used here. For retrieval systems the usable signal is uptake, as a detector of ignored evidence; sufficiency is tracked about as well by the model’s own abstention, and the abstention gate trades abstention against compliance uniformly rather than making the model more evidence-sensitive.

Limitations. Causal experiments on the external sources use Qwen only; RAGBench correctness relies on an LLM judge; Wikipedia-trained directions transfer only partially to other domains; uptake weakens on

3–4-hop document-disjoint splits (0.61); and only additive and projection interventions at the anchor were tested.

8 Related work

9 Conclusion

Paired activation updates over matched evidence trajectories show that evidence sufficiency is readable before generation, transferable and text-recoverable; that evidence uptake is readable, transferable and *not* text-recoverable; and that neither readout direction causes the behaviour it predicts, the one causal direction being an abstention gate. We release the trajectory construction, probing and intervention code so that the readout-versus-mechanism check becomes standard for evidence-aware generation.

References

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- [2] Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. Musique: Multihop questions via single-hop question composition. In *Transactions of the Association for Computational Linguistics*, 2022.

A Evidence conditions and increments

For a question with n gold hop paragraphs $g_0, \dots, g_{n-1}$ (canonical order from the dataset’s decomposition) and its distractor pool, the conditions are:

- C_0 : no evidence; C_D : n distractors, each length-matched (word count) to one gold paragraph.
- Chains: for each hop order $o \in \{\text{canonical}, \text{reverse}\} \cup \{\text{random permutations}, n \geq 3\}$, the prefixes $C_{o_1}, C_{o_1 o_2}, \dots$; each new paragraph is inserted at a uniformly random position in the previous presentation order. The last prefix is minimally sufficient. Identical conditions arising from different orders are de-duplicated.
- Leave-one-out ($n \geq 3$): all gold paragraphs but one, shuffled, and its completion.
- From every penultimate state S (used by a chain or leave-one-out context): $S+D$ (distractor length-matched to the decisive paragraph), $S+R$ (verbatim duplicate of a gold paragraph already in S), $S+F$ (decisive paragraph with its sub-answer, and title if applicable, replaced by another question’s sub-answer of the same relation type; questions whose sub-answer does not appear in the paragraph have no false condition), $S+A$ (a distractor with “(Not to be confused with {answer}.)” appended).
- Post-sufficiency: the canonical minimal sufficient context plus a distractor, plus a duplicate gold, plus a conflicting falsified final hop; and the full context of all 20 paragraphs, shuffled.

Increment kinds and labels: `decisive/loo_decisive` ($y^{\text{suff}} = 1$), `gold_nondecisive`, `distractor`, `redundant`, `false`, `answer_ctrl`, `distractor_from_none`, `post_distractor`, `post_redundant`, `post_false`, `full_context`. The *sufficiency* task uses all kinds; the *matched* variant uses decisive increments and the four controls from the same penultimate state only. For 2-hop questions this gives 17 conditions and 17 increments per question; the conversational protocol re-runs the 10 chain-related increment kinds with the model’s own stateless answer as A_k (30,000 increments for Qwen).

B Probing, nuisance and intervention details

Activation capture. Activations are read by forward hooks on each decoder block’s output at the anchor position, in a single left-padded batched prefill that also appends the gold answer tokens after the anchor (causality leaves the anchor unaffected) to obtain the teacher-forced gold log-probability; the KV cache is then cropped and greedy decoding continues. Layer index l denotes the residual stream after block l ($l = 0$: embeddings).

Probes. `scikit-learn` logistic regression, $C = 0.5$, balanced class weights, features standardised on the training split; the layer is selected by grouped 3-fold cross-validation on training questions (never on test), and the reported number is the test AUROC at that layer. Directions are w/σ normalised to unit norm in raw space. The difference-of-means direction is the normalised difference of class means of Δh (or of h for state tasks). Bootstrap intervals resample test questions (1,000 draws) using the stored probe weights.

Nuisance covariates. Δ tokens, total tokens, Δ gold log-probability, Δ next-token entropy, Δ top probability, answer string in added text / in the full context / in the previous context, TF-IDF cosine relevance of the added passage to the question, number and word count of added paragraphs, relative insertion position, and a 16-dimensional truncated-SVD embedding of the added text’s TF-IDF vector (fit on training increments). The residualised probe regresses Δh on these with ridge ($\lambda = 1$) and probes the residual. The “gain” adds out-of-fold (5-fold, grouped by question) probe scores to the nuisance classifier.

Text baselines. TF-IDF (1–2-grams, sublinear tf, $\leq 200k$ features) with logistic regression; DeBERTa-v3-base fine-tuned with AdamW (lr 3×10^{-5} , linear decay, class-weighted cross-entropy). The reduced budget uses one epoch, 256 tokens and 8,000 training increments (all positives kept); the full budget two epochs, 512 tokens and all training increments.

Interventions. Steering adds the vector at the anchor and every generated position (“anchor onward”) or at all positions; the unit u is $|\mathbb{E}[v^\top h \mid \text{sufficient}] - \mathbb{E}[v^\top h \mid \text{insufficient}]|$ on training conditions, computed per layer. Projection-out sets $v^\top h$ to the insufficient-state mean at the anchor and generated positions for every layer from the best layer onward. Patching replaces the anchor residual during the prefill at every layer from the best layer to the last; decoding then proceeds from the patched cache. Principal components are computed from 1,500 training decisive updates per layer; the “unembedding” direction is the normalised row of the output embedding matrix for the gold answer’s first token.

C Behaviour by condition family

Table 5 reports, for every condition family of Appendix A, the fraction of greedy answers that are correct and the fraction that are INSUFFICIENT, in the evidence-only regime over the 2,000 two-hop questions of each model. These are the behavioural labels from which uptake, correction, stability and revision are derived (Section 2).

Condition (correct / abstain)	Qwen2.5-7B	Llama-3.1-8B	Mistral-7B
no evidence	0.00 / 1.00	0.00 / 1.00	0.00 / 0.99
distractors only	0.00 / 0.97	0.00 / 0.97	0.00 / 0.94
one hop	0.10 / 0.80	0.12 / 0.76	0.13 / 0.69
minimal sufficient	0.46 / 0.30	0.47 / 0.31	0.36 / 0.36
one hop + false final hop	0.13 / 0.49	0.13 / 0.55	0.12 / 0.51
sufficient + distractor	0.39 / 0.35	0.38 / 0.41	0.27 / 0.47
full context (20 par.)	0.22 / 0.38	0.17 / 0.61	0.17 / 0.48

Table 5: Behaviour by condition family (evidence-only regime, 2,000 questions per model): fraction correct / fraction answering INSUFFICIENT.

D Robustness across data splits

Every probe is trained on 70% of questions and tested on the remaining 30% under four groupings that are never shared between train and test: question id; gold document (connected components of questions that share a supporting page); answer entity; and relation-type tuple. The document split is the default in the main text. Table 6 gives the best-layer test AUROC under all four for the three two-hop runs, the 3–4-hop run and 2WikiMultiHopQA. Sufficiency varies by at most 0.015 across splits. Uptake is lower on the document and answer splits than on the question split, by 0.06–0.13 on the two-hop runs and by 0.18 on 3–4-hop questions (Appendix H).

Run	Task	question	document	answer	relation
Qwen2.5-7B	Suff.	0.956	0.948	0.945	0.947
Qwen2.5-7B	Uptake	0.851	0.775	0.726	0.841
Llama-3.1-8B	Suff.	0.960	0.952	0.958	0.959
Llama-3.1-8B	Uptake	0.845	0.783	0.736	0.815
Mistral-7B	Suff.	0.964	0.949	0.954	0.958
Mistral-7B	Uptake	0.873	0.771	0.758	0.793
Qwen 3–4 hop	Suff.	0.931	0.925	0.923	0.932
Qwen 3–4 hop	Uptake	0.795	0.613	0.611	0.795
Qwen 2Wiki	Suff.	0.986	0.986	0.981	0.983
Qwen 2Wiki	Uptake	0.894	0.894	0.866	0.786

Table 6: Best-layer AUROC under the four disjoint splits, for the three 2-hop runs, the 3–4-hop run and 2WikiMultiHopQA.

E Evidence-augmented regime

Setup. The evidence-augmented regime repeats every stateless forward pass of the Qwen two-hop run with one change to the instruction: “use the supplied evidence together with your own knowledge” instead of “use only the supplied evidence”. Conditions, increments, splits and probes are otherwise identical.

Results. The Qwen probes give sufficiency 0.945, matched 0.913, uptake 0.721 and correction 0.786 (document split), within 0.05 of the evidence-only regime. Behaviour shifts as expected for a more permissive instruction: on minimally sufficient contexts the model answers more (0.51 correct, 0.23 abstain) and it follows false evidence more often.

F Cross-task and zero-retraining transfer

Cross-task transfer. For every pair of tasks we apply the direction learned for one task (row) to the increments of another (column), at the row task’s best layer, on the document split’s test questions (Fig. 5). Uptake and correction share a direction (cross-task AUROC 0.82), whereas the sufficiency direction reaches only 0.54–0.60 on uptake.

Zero-retraining transfer. Table 7 applies the Qwen two-hop stateless directions, unchanged and at the source’s best layer, to the conversational protocol (Appendix G), to 2WikiMultiHopQA (Appendix I) and to 3–4-hop MuSiQue (Appendix H). All target increments are scored, since none of them were used to train the source probe.



Figure 5: Cross-task transfer (Qwen2.5-7B, document split): AUROC of the direction learned for the row task, evaluated on the column task.

Direction (MuSiQue 2-hop, stateless)	→ conversational	→ 2WikiMultiHopQA	→ MuSiQue 3–4 hop
sufficiency	0.790	0.956	0.822
sufficiency matched	0.776	0.931	0.849
uptake	0.776	0.791	0.675
correction	0.920	0.846	0.744
stability	0.820	0.822	0.755
revision	0.659	0.770	0.708

Table 7: Zero-retraining transfer of Qwen2.5-7B 2-hop directions (AUROC at the source’s best layer).

G Conversational protocol

Setup. In the conversational revision protocol the model sees, in a two-turn chat, the question with its earlier evidence and its own stateless answer A_k from that context, and then only the new paragraph, with the instruction to change its answer only if the total evidence warrants it. The paired update is $h_{\text{conv}}(C_k \rightarrow C_{k+1}) - h(C_k)$. The 10 chain-related increment kinds are re-run in this format (30,000 increments for Qwen); labels are defined as in the stateless protocol.

Results. Probes trained within the conversational regime reach sufficiency 0.951, uptake 0.818, correction 0.922, stability 0.959 and revision 0.969 (document split). Stability and revision are much higher than in the stateless regime because the model’s previous answer is in the prompt. The stateless directions transfer at 0.79 (sufficiency), 0.78 (uptake), 0.92 (correction) and 0.82 (stability) (Table 7).

H 3–4-hop questions

Setup. A separate Qwen run uses 3- and 4-hop MuSiQue training questions (closed-book accuracy 1.3%), with the condition set of Appendix A including random hop orders and leave-one-out contexts, which exist only for $n \geq 3$.

Results. Sufficiency probes reach 0.931/0.925/0.923/0.932 (question/document/answer/relation split); matched 0.907/0.900/0.896/0.906; uptake 0.795/0.613/0.611/0.795. Along the sufficiency direction, chain-decisive and leave-one-out-decisive increments have the same mean projection (+0.023), followed by false (+0.003), distractor (−0.02), redundant and answer-string (−0.06) and non-decisive gold increments (−0.11). Patching

(correct/answered): none 0.05/0.20, full 0.25/0.62, parallel 0.06/0.19, orthogonal 0.25/0.62, PC256 0.20/0.51, random-256 0.06/0.20; projecting out v_{dm} leaves 0.03 answered. The two-hop directions transfer at 0.82–0.85 for sufficiency (Table 7).

I 2WikiMultiHopQA

Setup. The Qwen pipeline is run unchanged on 2WikiMultiHopQA two-hop questions, with gold paragraphs from its supporting facts and its own distractors.

Results. Within-dataset probes (document split): sufficiency 0.986, matched 0.972, uptake 0.894, correction 0.914. Here the matched task is almost fully explained by nuisance covariates (0.965, chiefly the change in gold-answer log-probability); the residualised probe reaches 0.82. The MuSiQue directions transfer at 0.96 (sufficiency), 0.93 (matched), 0.85 (correction) and 0.79 (uptake) (Table 7).

J Closed-book transfer

Setup. We re-run the closed-book prompt on Qwen’s screened pool of 4,000 questions with activation capture and score every stored direction on the closed-book anchor activations, for two targets: closed-book correctness and closed-book abstention (INSUFFICIENT). A probe trained directly on closed-book activations (question-disjoint split) gives the ceiling.

Results. Closed-book answers are correct 1.5% of the time and abstain 91.7% of the time. For correctness, the state-correctness direction reaches 0.810 (0.838 at its best layer), correction 0.799 and state-sufficiency 0.722 (0.808), against an own-probe ceiling of 0.874. The abstention-flag direction predicts abstention at 0.977, against a ceiling of 0.973.

K Steering along the logistic directions

Setup. We add αuv along the logistic state-sufficiency direction at its best layer, from the anchor onward, with u the sufficient-minus-insufficient gap along v (Appendix B). Doses are $\alpha \in \{\pm 0.5, \pm 1, \pm 2\}$ gap units ($\leq 1.4 \ell_2$) and $\alpha \in \{\pm 8, \pm 16, 32\}$ (5–22 ℓ_2 , on a mean residual norm of 61). We also steer along the uptake direction at layer 26 ($\leq 16 \ell_2$), and run an all-positions variant. A random unit direction with the same ℓ_2 magnitude is the control. Outcomes are measured on 150 held-out questions in each of five sets: sufficient, insufficient, false-evidence, answer-string and distractor-only contexts.

Results. Every rate stays within ± 0.02 of baseline on all five sets, for the probe directions and the random control alike. At 32 units all rates degrade by 0.02–0.05, again equally for probe and random directions. The all-positions variant at $\leq 11 \ell_2$ is equally flat.

L Difference-of-means steering

Low dose (main protocol). We add $\alpha u_l v_{\text{dm}}^{(l)}$ at four layers from the best sufficiency layer, from the anchor onward, with $\alpha \in \{\pm 0.5, \pm 1\}$ in units of the per-layer sufficient-minus-insufficient gap, on 150 held-out questions per set. Table 8 gives the full numbers behind Fig. 4.

High dose. Adding $\alpha u_l v_{\text{dm}}^{(l)}$ at all positions and at every layer from the best layer onward with $|\alpha| \geq 2$ (per-layer gaps of 5–14 ℓ_2) produces degenerate output: every prompt is “answered”, never correctly and never with INSUFFICIENT. This happens equally for the probe direction and the random control, and is why the main protocol uses four layers, anchor onward and $|\alpha| \leq 1$.

Model	α	sufficient corr./ans.	insufficient ans.	distractors ans.	false ans./followed	answer-string ans.
Qwen2.5-7B	-1.0	0.23/0.34	0.05	0.00	0.23/0.06	0.03
	-0.5	0.34/0.52	0.09	0.01	0.43/0.13	0.09
	+0.5	0.49/0.79	0.20	0.06	0.75/0.25	0.29
	+1.0	0.49/0.82	0.26	0.12	0.79/0.25	0.35
Llama-3.1-8B	-1.0	0.03/0.07	0.00	0.00	0.03/0.01	0.00
	-0.5	0.25/0.38	0.07	0.00	0.23/0.12	0.05
	+0.5	0.65/0.89	0.27	0.13	0.82/0.35	0.34
	+1.0	0.69/0.94	0.51	0.29	0.92/0.39	0.59
Mistral-7B	-1.0	0.01/0.08	0.06	0.11	0.06/0.01	0.04
	-0.5	0.11/0.33	0.09	0.01	0.25/0.09	0.04
	+0.5	0.47/0.78	0.31	0.12	0.75/0.29	0.29
	+1.0	0.54/0.87	0.47	0.28	0.87/0.28	0.43

Table 8: Low-dose steering along v_{dm} (150 held-out questions per set). Baselines ($\alpha = 0$) appear in Fig. 4.

M Single-layer patching

Setup. As in Section 5, the insufficient prompt’s anchor residual is replaced by the sufficient run’s (full patch), but at a single layer only instead of every layer from the best one onward.

Results. Patching only the best sufficiency layer (16 for Qwen) leaves the insufficient prompt unchanged in all variants. Patching only layer 26 with the full sufficient residual restores 0.18/0.62 (correct/answered), so late-layer anchor content alone suffices to make the model answer.

N Adaptive retrieval

The readouts of Section 4 are computed before generation, so they can drive retrieval: after each round, read the anchor and decide whether to stop and answer or to retrieve more. We test every learned direction in this role, both as a *stopping rule* and as a *verifier* of the answer given at the stopping round.

Simulation. Each held-out question is walked along its canonical trajectory, one retrieval round per stage: (1) the matched distractors, (2) the first gold hop, (3) the minimal sufficient context, (4) the sufficient context plus a distractor. A policy that never stops answers from stage 4; an oracle stops at stage 3. At round k a policy reads either the state $h(C_k)$ (state probes) or the update the round produced, $\Delta h = h(C_k) - h(C_{\text{prev}})$ (update probes), projects it on the direction at the direction’s best layer, and stops when the projection crosses a threshold. The threshold *and the sign* are chosen on training questions to maximise accuracy at the stopping stage (ties broken by fewer rounds), so each direction is used in whichever reading serves retrieval best. For stability and revision we also report the reading that stops when the probe says the last round *changed the answer*. Baselines: never stop; the oracle; the behavioural rule, which stops at the first round whose greedy answer is not INSUFFICIENT (it needs a generation per round, which the probes do not); and a next-token-entropy threshold calibrated the same way. Accuracy counts abstentions as wrong. Everything is computed from stored activations and behaviour, with no new forward passes. Thresholds are calibrated on the questions the probes were trained on, which makes the probe policies, if anything, slightly over-confident on test questions. The trajectories present gold hops in order, so the simulation measures *when to stop*, not what to retrieve.

Stopping rules (Table 9, Fig. 6). Only the sufficiency signals are useful stopping rules. The update-sufficiency v_{dm} direction is the best learned rule on all three MuSiQue models (0.478, 0.468, 0.336). It matches the oracle (0.475, 0.470, 0.336) in about three rounds, and on Qwen it is the one rule significantly better than the behavioural rule. The logistic state and update sufficiency probes are statistically tied with the behavioural rule on MuSiQue, while acting before any generation. On HotpotQA both beat it (0.732 and

Stopping rule	MuSiQue Qwen	MuSiQue Llama	MuSiQue Mistral	HotpotQA Qwen	2WikiMultiHopQA Qwen	Synthetic Qwen
<i>Baselines</i>						
full context (never stop)	0.404 (4.00)	0.385 (4.00)	0.243 (4.00)	0.685 (3.99) ⁺	0.460 (4.00)	0.966 (3.65) ⁺
oracle (stop when sufficient)	0.475 (3.00)	0.470 (3.00)	0.336 (3.00)	0.740 (3.00) ⁺	0.583 (3.00)	0.973 (2.93) ⁺
behavioural (first non-INSUFFICIENT)	0.460 (3.09)	0.460 (3.09)	0.318 (3.08)	0.650 (2.53)	0.593 (3.10)	0.939 (2.88)
next-token entropy	0.381 (3.85) ⁻	0.378 (3.96)	0.261 (3.91)	0.690 (3.74) ⁺	0.467 (3.67) ⁻	0.969 (3.35) ⁺
<i>State probes, $h(C_k)$</i>						
sufficiency (logistic)	0.466 (3.13)	0.448 (3.11)	0.326 (3.13)	0.732 (3.08)⁺	0.583 (3.02)	0.977 (2.96)⁺
sufficiency (v_{dm})	0.407 (3.94)	0.385 (3.58)	0.261 (3.63)	0.690 (3.86) ⁺	0.463 (3.71)	0.967 (3.00) ⁺
correctness	0.399 (3.57)	0.443 (3.51)	0.284 (3.64)	0.667 (2.92)	0.547 (3.42)	0.964 (3.23) ⁺
abstention flag	0.435 (3.07) ⁻	0.435 (3.12) ⁻	0.302 (3.13) ⁻	0.702 (3.18) ⁺	0.583 (3.12) ⁻	0.969 (2.99) ⁻
<i>Update probes, Δh of the round</i>						
sufficiency (logistic)	0.458 (3.30)	0.460 (3.09)	0.316 (3.18)	0.728 (3.23) ⁺	0.570 (3.16)	0.949 (3.32)
sufficiency (v_{dm})	0.478 (3.06)⁺	0.468 (2.98)	0.336 (3.00)	0.727 (3.07) ⁺	0.543 (3.32)	0.970 (3.15) ⁺
sufficiency matched	0.404 (4.00)	0.385 (4.00) ⁻	0.243 (4.00)	0.683 (3.98) ⁺	0.513 (3.21)	0.858 (3.22)
uptake	0.417 (3.55)	0.388 (3.35)	0.259 (3.47)	0.683 (3.98)	0.507 (3.42)	0.957 (3.07) ⁺
correction	0.404 (3.43)	0.405 (3.38)	0.302 (3.53)	0.667 (2.93)	0.550 (3.36)	0.920 (3.19)
stability	0.407 (3.46)	0.400 (3.42)	0.243 (4.00) ⁻	0.685 (3.99) ⁺	0.543 (3.38)	0.964 (3.63) ⁺
stability, stop when the answer changes	0.402 (3.99)	0.385 (4.00)	0.243 (4.00)	0.685 (3.99)	0.457 (3.99)	0.966 (3.65)
revision (stop when the answer changes)	0.396 (3.13)	0.363 (3.12)	0.236 (3.23)	0.690 (3.90) ⁺	0.513 (3.07)	0.934 (3.02)
test questions	609	600	610	600	300	830

Table 9: Every direction as a stopping rule: accuracy at the stopping round (mean rounds in parentheses), held-out questions of the document split. Bold: best non-oracle rule per column. ⁻: the training-optimal reading stops when the projection is *low*. ⁺: the 95% paired bootstrap interval of the accuracy difference to the behavioural rule lies above zero (1,000 resamples of test questions).

0.728 vs 0.650), and on the synthetic data the state probe does (0.977 vs 0.939), because the behavioural rule stops too early there (HotpotQA: 2.53 rounds). On 2WikiMultiHopQA they tie it (0.583 vs 0.593). The other directions are weak stopping rules on every source. Uptake, correction and stability are trained on decisive or already-correct increments and say nothing about whether the evidence is complete. Their best thresholds stay within 0.06 of the never-stop policy on MuSiQue (0.24–0.42) and below the best sufficiency rule on every source. Read as “stop when the answer changes”, stability never fires (about 4.0 rounds, at full-context accuracy). Revision, the direct change detector, stops at about three rounds but often on the first wrong answer, and is the worst learned rule on MuSiQue (0.236–0.396). The v_{dm} direction of the *state*, which the causal analysis identified as an abstention gate (Section 5), is a poor stopping rule (0.26–0.41 on MuSiQue). Its update counterpart is the best one. Figure 6 sweeps each threshold on test questions: the sufficiency curves peak at the oracle’s operating point, while the uptake and stability curves rise roughly monotonically towards the full-context accuracy and never cross the behavioural rule on MuSiQue.

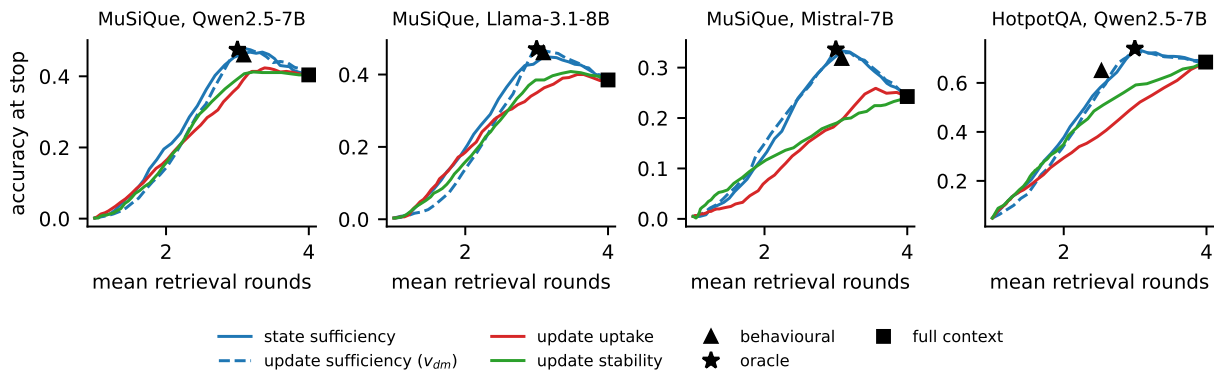


Figure 6: Accuracy against mean retrieval rounds when each direction’s stopping threshold is swept over test questions (41 quantiles), with the behavioural rule, oracle and full context as points.

Verification at the stopping round (Table 10). The directions that fail as stopping rules are useful once retrieval has stopped. We take the stopping round chosen by the state-sufficiency rule and ask, among the questions the model answers there, how well each signal ranks correct above wrong answers. This is the decision to output, flag or re-retrieve. The uptake direction is the best learned verifier on MuSiQue

Qwen (0.672) and 2WikiMultiHopQA (0.847), where it beats output confidence (top-token probability 0.621, 0.705). Correction is the best learned verifier on Mistral (0.666 vs 0.630). By AUROC, next-token entropy is ahead on Llama and HotpotQA (by 0.02) and on the synthetic data (by 0.11, where 99% of answered stops are correct). For selective answering, keeping only the more confident half of the answers, uptake gives higher accuracy than top-token probability in five of six settings (0.789 vs 0.751 on MuSiQue Qwen, 0.957 vs 0.870 on 2WikiMultiHopQA), and is level on the synthetic data (0.995 vs 0.998). The sufficiency directions are near chance as verifiers on MuSiQue and HotpotQA (0.46–0.60), and the abstention flag is below chance everywhere. They detect completeness, which the stopping rule has already selected for, not whether the answer is right. This division of labour mirrors the readout results. Sufficiency tells the system when the evidence is complete. Uptake, nearly orthogonal to it (Section 4), tells whether the model has actually integrated that evidence into its answer.

Signal at the stopping round	MuSiQue Qwen	MuSiQue Llama	MuSiQue Mistral	HotpotQA Qwen	2WikiMultiHopQA Qwen	Synthetic Qwen
top-token probability	0.621	0.615	0.630	0.654	0.705	0.876
–next-token entropy	0.623	0.630	0.637	0.657	0.705	0.877
state: sufficiency	0.465	0.463	0.521	0.526	0.621	0.738
state: correctness	0.560	0.568	0.618	0.626	0.763	0.388
state: abstention flag	0.432	0.473	0.412	0.395	0.444	0.252
update: sufficiency	0.482	0.543	0.495	0.554	0.594	0.585
update: sufficiency (v_{dim})	0.566	0.569	0.601	0.567	0.806	0.534
update: uptake	0.672	0.613	0.647	0.617	0.847	0.764
update: correction	0.594	0.576	0.666	0.639	0.753	0.400
update: stability	0.549	0.566	0.559	0.562	0.775	0.497
update: revision	0.512	0.517	0.490	0.524	0.601	0.531
accuracy of all answered stops	0.665	0.666	0.534	0.810	0.761	0.990
accuracy, top half by top-token prob.	0.751	0.733	0.618	0.863	0.870	0.998
accuracy, top half by uptake	0.789	0.757	0.651	0.867	0.957	0.995
answered stops	427	404	373	542	230	819

Table 10: Verification once retrieval stops (state-sufficiency stopping rule): AUROC of each signal at the stopping round for the correctness of the answer given there (answered stops only). Bold: best per column. Bottom: accuracy of all answered stops and of the half ranked highest by top-token probability or by the uptake direction.

Other sources. On 3–4-hop MuSiQue the trajectories end at the sufficient context (no post-sufficiency stage), so the oracle and the never-stop policy coincide (0.360). All learned rules reach this value, while the behavioural rule stops too early (0.340). On RAGBench (relation split, 955 test questions) most trajectories have two or three stages and no distractor after sufficiency, and every policy lies within 0.21–0.23. With the original protocol of a threshold at 0.97 training precision, the state-sufficiency rule reached 0.442 in 2.87 rounds on MuSiQue Qwen and 0.707 in 2.90 on HotpotQA. Accuracy-optimal calibration trades a few rounds for accuracy on MuSiQue.

O Results by question group on the external sources

Table 11 breaks down the held-out probe AUROC of the external sources (Section 6) by question group, to separate the evidence-state readout from properties of the questions. The groups are: closed-book status (HotpotQA and RAGBench keep closed-book-correct questions, `select: all`); question type (HotpotQA bridge versus comparison, whose two facts are independent lookups); domain (RAGBench’s twelve sub-datasets grouped into biomedical, finance, technical, customer support, general web QA and Wikipedia); and hop count (synthetic 1–3 hops; for RAGBench, one versus two gold documents). The stored best-layer probe of each source, trained on all groups, is scored per group on test questions without refitting; increment counts are in parentheses.

Dataset	grouping	group	sufficiency	uptake	correction
HotpotQA	closed book	incorrect	0.978 (7840)	0.724 (986)	0.839 (5554)
HotpotQA	closed book	correct	0.984 (1689)	0.802 (214)	0.885 (760)
HotpotQA	qtype	bridge	0.978 (7878)	0.680 (982)	0.839 (4963)
HotpotQA	qtype	comparison	0.985 (1651)	0.787 (218)	0.880 (1351)
RAGBench	domain	biomedical	0.929 (1473)	0.776 (291)	0.818 (1331)
RAGBench	domain	finance	0.899 (1081)	0.641 (206)	0.769 (949)
RAGBench	domain	technical	0.881 (1069)	0.675 (196)	0.733 (989)
RAGBench	domain	customer support	0.897 (716)	0.569 (153)	0.658 (587)
RAGBench	domain	general	0.835 (1821)	0.721 (352)	0.702 (1520)
RAGBench	domain	wikipedia	0.978 (729)	0.557 (158)	0.766 (612)
RAGBench	n hops	1	0.836 (2476)	0.698 (444)	0.785 (2211)
RAGBench	n hops	2	0.931 (4413)	0.677 (912)	0.711 (3777)
Synthetic	n hops	1	0.922 (3645)	0.992 (486)	0.875 (2450)
Synthetic	n hops	2	0.970 (6192)	0.928 (1032)	0.926 (4929)
Synthetic	n hops	3	0.970 (11501)	0.916 (2395)	0.931 (8859)

Table 11: Held-out probe AUROC by question group (test-set increment counts in parentheses): closed-book status, question type, domain and hop count.

P The Evidence State Transition Benchmark

The Evidence State Transition Benchmark (ESTB) tests whether the evidence-state readout and its causal profile, established on MuSiQue trajectories in Sections 4 and 5, survive controlled changes to the evidence. It puts all of our sources under one labelling scheme and one evaluation protocol. This appendix describes what the benchmark measures, how each axis was built, and the full results of the retrieval-source and linguistic-form runs, which appear only in summary in Section 6.

P.1 What the benchmark measures

Unit of measurement. As in the rest of the paper, the unit is an *evidence state transition*: an increment $C_{k-1} \rightarrow C_k$ that adds one paragraph (or, for post-sufficiency and full-context increments, a set of paragraphs) to the context, with the paired update $\Delta h = h(C_k) - h(C_{k-1})$ read at the answer-start anchor. ESTB annotates every increment with the *quality state* of the evidence it adds (Table 12) and with the values of the other axes (Table 13): retrieval source, linguistic form, domain, question type, hop count and closed-book status.

Labels. Each increment carries the sufficiency label $y^{\text{suff}} = \mathbf{1}[C_k \text{ sufficient}, C_{k-1} \text{ not}]$, which is defined by construction from the gold hops and does not depend on the model, and a behavioural uptake label for increments that cross sufficiency (answer becomes correct or not). An increment crosses sufficiency exactly when it supplies the last missing hop, whether in the original wording (C3), paraphrased, or from an independently authored source.

Measurements per cell. A *cell* is the set of increments of one run that share one value on one axis (for example, “E5-retrieved distractors, quality state C1”). For every cell we report:

1. **Within-run readout:** the test AUROC of the run’s own held-out probe (trained on the run’s training questions), for sufficiency, matched sufficiency and uptake. This is the ceiling: is the state linearly decodable at all under this kind of evidence?
2. **Transfer without retraining:** the AUROC of the Qwen MuSiQue two-hop direction (document split, at its best layer), applied unchanged. Because the ESTB runs sample from the same MuSiQue pool as the source run, 818 of the 1,200 questions in the benchmark-gold runs (273 of 402 and 152 of 219 in the fully retrieved runs) are in the source probe’s training split. These questions are *excluded* from every transfer score, so transfer is measured only on questions the source probe never saw.
3. **Quality-state separability:** for each quality state Q , the AUROC separating Q ’s increments (negatives) from the decisive increments of the same run (positives). Near 1, the readout cleanly distinguishes Q from completing the evidence; lower values mean that Q looks like sufficiency to the probe.

4. **Causal profile:** on 150 held-out questions, the answered rate of the insufficient prompt unpatched, with the full sufficient anchor residual patched in, and with only the probe-direction component ($\parallel v$) patched in, all at every layer from the best one onward; and the answered rate of sufficient prompts after projecting v_{dm} to its insufficient-state value (Section 5).

The benchmark therefore separates three questions: whether the state is *present* (1), whether it is the *same* state as on MuSiQue (2, 3), and whether its causal profile, an inert probe direction alongside a causally sufficient anchor and a v_{dm} abstention gate, carries over (4).

What the benchmark does not measure. ESTB does not score answer quality or retrieval quality for their own sake: retrieval and paraphrasing only vary the evidence that enters a trajectory. It does not ask whether the model *should* answer from partial or contradictory evidence; those states are scored by whether the readout separates them from genuine completion. The sufficiency label is structural (all hops present) rather than semantic, so a passage that happens to state a fact in a way our string-based labelling misses (Appendix P.7) is counted as insufficient.

P.2 Evidence quality states

Table 12 lists the seven quality states of the review’s evidence-quality axis and the increment kinds (Appendix A) that realise them. All states except C0 are increments, i.e. transitions *into* the state from the preceding context.

State	Name	Increment kinds (added evidence)	y^{suff}
C0	none	no evidence (start state of every trajectory)	–
C1	irrelevant	distractor_from_none , distractor : a length-matched distractor added to an empty or penultimate context	0
C2	partial	gold_nondecisive : a gold hop that leaves another hop missing; partial : the decisive paragraph rewritten to withhold its value; redundant from a penultimate state (C2+redundant)	0
C3	complete	decisive , loo_decisive : the last missing hop; decisive_paraphrase , decisive_independent : the same hop paraphrased or from an independent source	1
C4	complete + redundant	post_distractor , post_redundant , post_redundant_paraphrase , post_redundant_independent , full_context : further evidence added after sufficiency	0
C5	contradictory	false : the decisive hop with its value replaced (from a penultimate state); post_false : a falsified copy added to a sufficient context	0
C6	answer string, no relation	answer_ctrl : a distractor with “(Not to be confused with {answer}.)” appended	0

Table 12: ESTB evidence-quality states and the increment kinds that realise them. y^{suff} : whether the increment crosses into sufficiency.

The contradictory state has two sources. On benchmark and retrieved trajectories, the false value is another question’s sub-answer of the same relation type (**false_synthetic**); on the synthetic benchmark, it is a generated value of the same type. The tables label the first kind “C5 contradictory (synthetic)” because the false value is substituted rather than found in a real document; naturally occurring contradictions are not separated from **near_miss** passages by our labelling (Appendix P.7). Partial evidence is defined in two ways: for multi-hop questions, a gold hop that leaves another hop missing (“partial (hop)”); for any question, the decisive paragraph rewritten so that its subject and relation remain but the value is withheld (“partial (masked)”, Appendix P.5).

P.3 Axes and sources

All ESTB runs reported here use Qwen2.5-7B-Instruct and the pipeline of the main text without modification. The four retrieval runs and the variants run draw 1,200 two-hop MuSiQue training questions that the model answers incorrectly closed-book, with no random hop permutations.

Axis	Levels	Source (runs)
Evidence quality	C0–C6 (Table 12)	every run
Retrieval source	benchmark; BM25; E5 (dense); each as distractors only or as all evidence	MuSiQue 2-hop rebuilt over wiki-18 (Appendix P.4)
Linguistic form	verbatim; LLM paraphrase; independently authored; templated (5 families)	MuSiQue variants run (Appendix P.5); synthetic benchmark
Domain	Wikipedia; biomedical; finance; technical; customer support; general web QA	MuSiQue, HotpotQA, 2WikiMultiHopQA; RAGBench’s 12 sub-datasets
Question type	bridge; comparison; single-hop	HotpotQA; RAGBench; synthetic
Hop count	1; 2; 3–4	synthetic, RAGBench; MuSiQue 2-hop; MuSiQue 3–4-hop
Closed-book status	correct; incorrect	HotpotQA and RAGBench (<code>select: all</code>)

Table 13: ESTB axes. The domain, question-type, hop-count and closed-book axes reuse the external runs of Section 6, scored per group in Appendix O; the retrieval-source and linguistic-form axes required new construction and are detailed here.

P.4 Retrieval-source axis

Corpus and retrievers. Evidence comes from wiki-18, the December 2018 Wikipedia dump released with FlashRAG: 21M passages of 100 words with their article titles. *BM25* uses `bm25s` over title and text, with English stop-words and Snowball stemming. *E5* uses the published FAISS flat inner-product index of `e5-base-v2` passage embeddings; queries are encoded with the `query: prefix` and mean pooling, then ℓ_2 -normalised.

Queries. A single question-level query rarely retrieves both hops of a bridge question. Each question therefore issues the question itself plus one query per hop, taken from MuSiQue’s decomposition with the earlier hops’ sub-answers substituted for the `#k` placeholders (three queries per two-hop question). This imitates an iterative or decomposing RAG system that gets the earlier hops right. The per-query hit lists are merged round-robin by rank, de-duplicated, and cut at $k = 30$ passages per question.

Passage labels. Each retrieved passage is labelled against the question’s gold hops by title and string matching (titles normalised by lower-casing and removing a trailing parenthetical):

- **retrieved_gold:** the title matches a gold hop’s page *and* the passage contains that hop’s sub-answer;
- **same_page:** the title matches a gold hop’s page but the sub-answer is absent, i.e. partial evidence found in the wild;
- **near_miss:** not a gold page, but the passage mentions a bridge entity or the final answer, which is the natural analogue of the answer-string control C6;
- **irrelevant:** everything else.

Table 14 gives the label composition. E5 retrieves at least one gold passage for 85.5% of questions and both hops for 33.5%; BM25 does so for 68.1% and 18.2%.

Trajectories. Two modes turn retrieval output into trajectories.

- **Retrieved distractors** (*noise* mode): the benchmark gold paragraphs are kept and every distractor is replaced by a retrieved non-gold passage. The 18 distractors per question are taken hardest first: same-page passages, then near misses, then irrelevant passages, each in rank order. The trajectory builder then draws length-matched distractors from this pool exactly as it does from the benchmark pool. Distractors are thus no longer the benchmark’s curated negatives but what a retriever returns for the same question, and nearly a quarter of E5’s distractors come from the gold pages themselves.

Retriever	retrieved passages (top 30)				questions with gold for	
	gold	same page	near miss	irrelevant	≥ 1 hop	both hops
E5	0.080	0.139	0.292	0.489	0.855	0.335
BM25	0.053	0.080	0.331	0.535	0.681	0.182
Distractors used in the trajectories (18 per question): same page / near miss / irrelevant						
E5	0.231 / 0.460 / 0.309					
BM25	0.133 / 0.535 / 0.332					

Table 14: Retrieved passages by label (1,200 MuSiQue two-hop questions, multi-query retrieval, $k = 30$).

- **All evidence retrieved** (*full* mode): the gold paragraph of each hop is also replaced, by the highest-ranked `retrieved_gold` passage for that hop, so every paragraph the model sees is a 100-word wiki-18 passage. Only questions with a retrieved gold passage for both hops can be used: 402 for E5 and 219 for BM25.

Every other part of the construction (chains, controls, falsification, answer-string hatnotes, post-sufficiency and full-context increments) is unchanged, which yields the full set of quality states over retrieved evidence: 20,400 increments in each noise run, 6,809 (E5) and 3,703 (BM25) in the full runs.

P.5 Linguistic-form axis

LLM paraphrase and partial variants. For every gold hop paragraph of the 1,200 questions, Qwen2.5-7B-Instruct generates two rewrites by greedy decoding (at most 220 tokens):

- *paraphrase*: “Rewrite the following encyclopedia paragraph in completely different wording and sentence structure, keeping every fact exactly the same. You must keep the exact string ‘{sub-answer}’ verbatim.”
- *partial*: “Rewrite the following encyclopedia paragraph so that it still talks about ‘{subject}’ and the same topic, but withholds the specific fact ‘{sub-answer}’: do not mention ‘{sub-answer}’ or any equivalent name or value anywhere; instead say that the detail is not recorded here.”

Generations are validated before use. A paraphrase is kept only if it contains the sub-answer, is between $0.5\times$ and $2\times$ the original’s length, and shares fewer than 60% of its word 4-grams with the original. A partial is kept only if it contains neither the sub-answer, nor the final answer, nor any answer alias, has at least 15 words, and still mentions the hop’s subject. Of 2,400 hop paragraphs, 2,180 (90.8%) yield a valid paraphrase and 1,411 (58.8%) a valid partial.

Independently authored sources. MuSiQue, HotpotQA and 2WikiMultiHopQA all take paragraphs from Wikipedia, but from different dumps, with different paragraph boundaries and often different revisions. For each MuSiQue gold hop with page title T and sub-answer a , we search the gold and distractor paragraphs of the *other* two datasets for one whose normalised title is T , that contains a , and that shares fewer than 60% of its 4-grams with the MuSiQue paragraph; when several qualify, the least overlapping is used. This finds an independently written statement of the same relation for 228 of the 2,400 hops (144 from HotpotQA, 84 from 2WikiMultiHopQA).

Increments. From the penultimate state, the paraphrased and independent paragraphs give `decisive_paraphrase` and `decisive_independent` increments (C3, crossing sufficiency), and the partial paragraph gives a `partial` increment (C2, which does not). After sufficiency, the paraphrase or independent version of a hop already present gives `post_redundant_paraphrase` and `post_redundant_independent` increments (C4). Together with the standard kinds, the variants run has 25,632 increments, including 2,180 paraphrased decisive, 228 independent decisive and 1,411 partial ones.

Synthetic templated form. The synthetic benchmark (Section 6) supplies the fully controlled end of the axis: fictional entities over seven relations (founder, birthplace, capital, spouse, award, employer, headquarters), five template families per relation, one to three hops, and for every gold fact a paraphrase, a partial (value withheld) and a contradictory (different value) variant, with near-miss distractors (same relation, other subject; same subject, other relation) and unrelated filler.

P.6 Results

Evidence source	n_q	within-run probe (AUROC)			MuSiQue dir.		answered rate				
		suff.	matched	uptake	suff.	uptake	unpatched	full	$\parallel v$	proj.-out	v_{dm}
Benchmark gold + benchmark distractors	2000	0.948	0.911	0.775	(source)	(source)	0.13	0.71	0.13		0.01
Gold + E5-retrieved distractors	1200	0.946	0.909	0.681	0.935	0.817	0.10	0.77	0.10		0.02
Gold + BM25-retrieved distractors	1200	0.951	0.915	0.679	0.939	0.817	0.10	0.77	0.10		0.01
All evidence E5-retrieved	402	0.922	0.891	0.839	0.905	0.762	0.19	0.72	0.19		0.03
All evidence BM25-retrieved	219	0.912	0.876	0.748	0.910	0.820	0.30	0.68	0.30		0.02
+ LLM paraphrase / partial / independent	1200	0.935	0.885	0.718	0.929	0.796	0.10	0.77	0.10		0.00

Table 15: Retrieval-source and linguistic-form axes (Qwen2.5-7B, MuSiQue 2-hop). Within-run probes on the document split, except the two fully retrieved runs (question split, too few questions for document components). MuSiQue direction: transfer without retraining, scored only on questions outside the source probe’s training split. Answered rate: the insufficient prompt unpatched, with the full sufficient anchor, and with only the probe-direction component; and the sufficient prompt with v_{dm} projected to its insufficient value.

Retrieval source. With retrieved distractors (Table 15), within-run sufficiency is unchanged (0.946 E5, 0.951 BM25, against 0.948 with benchmark distractors), and the MuSiQue direction transfers at 0.935/0.939 on source-unseen questions. Retrieved distractors are harder than the benchmark’s: the transferred direction separates C1 increments from decisive ones at 0.906 (E5) and 0.917 (BM25), against 0.945 for benchmark distractors, and answer-string controls at 0.962/0.971 against 0.982. The run’s own probe recovers most of this (0.972/0.982 for C1), so the drop is a shift in the distractor distribution rather than a loss of the state. Uptake falls within run (0.68, against 0.78), but the transferred MuSiQue uptake direction scores 0.817 on the same increments: these runs have 1,200 rather than 2,000 questions, so the within-run probe is trained on fewer decisive increments, which is the likely reason. With all evidence retrieved, sufficiency stays at 0.91–0.92 within run and 0.905–0.910 transferred, and C1 separability is the lowest of any run (0.84 transferred, 0.89–0.92 own), which is expected when every paragraph is a 100-word passage from the same corpus as the gold ones. In every retrieval cell the causal profile of Section 5 reproduces: the full anchor patch raises answering from 0.10–0.30 to 0.68–0.77, the probe-direction component leaves it exactly at the unpatched rate, and projecting out v_{dm} leaves 0.01–0.03 answered.

Linguistic form. Paraphrased and independently authored decisive paragraphs are read as completing the evidence just as the original wording is. Against all non-decisive increments, the transferred MuSiQue direction scores original decisive increments at 0.931 AUROC, paraphrased ones at 0.929 and independently authored ones at 0.918, with mean projections 0.101, 0.098 and 0.096 (non-decisive: -0.082). Paraphrastic and independent redundancy after sufficiency is cleanly separated from completion (0.974 and 0.995 own probe, Table 16). The two hardest quality states are the ones that most resemble completion. Contradictory evidence added from the penultimate state (C5, the decisive paragraph with a swapped value) reaches only 0.76 own and 0.73 transferred, and it has the same “supplies the missing relation” surface form as the decisive paragraph. Masked partial evidence (C2), with subject and relation present but the value withheld, reaches 0.81 own and 0.74 transferred; its mean projection (0.044) lies between the non-decisive and decisive means. Partial evidence in the multi-hop sense (a non-decisive hop) is separated almost perfectly (1.00). Contradictory evidence is the hardest control in every retrieval run as well (0.71–0.75 transferred) and is also the state that projects closest to completion on the synthetic benchmark (Section 6).

Quality state vs. decisive increment	n	own probe	MuSiQue dir.
C5 contradictory (synthetic)	764	0.762	0.725
C2 partial (masked)	436	0.809	0.736
C1 irrelevant	1146	0.942	0.941
C4 complete+paraphrase	381	0.974	0.952
C6 answer string, no relation	764	0.975	0.979
C2+redundant	764	0.982	0.978
C4 complete+independent	57	0.995	0.950
C4 complete+irrelevant	382	0.997	0.980
C4 complete+redundant	382	0.999	0.984
C5 complete+contradictory	382	1.000	0.994
C2 partial (hop)	764	1.000	0.996
C4 full context	382	1.000	0.999

Table 16: Quality-state separability (MuSiQue variants run): AUROC separating each state’s increments (negatives) from decisive increments (positives), with the run’s own probe (test questions) and with the MuSiQue direction (source-unseen questions). Lower values mean the state looks more like completing the evidence.

Summary. Across the retrieval-source and linguistic-form axes, the sufficiency state remains decodable (≥ 0.91), the MuSiQue direction transfers to unseen questions (≥ 0.90), wording and authorship of the decisive evidence do not matter, and the causal dissociation reproduces in every cell. What changes is the difficulty of the controls: retrieved distractors from the gold pages and value-withheld or value-swapped versions of the decisive paragraph are where a deployed sufficiency detector would err.

P.7 Limitations of the construction

- **Oracle decomposition.** The per-hop retrieval queries substitute gold sub-answers into MuSiQue’s decomposition, so the retriever behaves like an iterative system that solves the earlier hops correctly. Gold coverage is an upper bound on what a realistic multi-hop retriever would achieve.
- **String-based passage labels.** `retrieved_gold` requires the gold page title and the sub-answer string. A passage on another page that states the same relation is labelled `near_miss` and used as a distractor; a gold-page passage that mentions the sub-answer in an unrelated sentence counts as gold. Both introduce some label noise, in opposite directions.
- **Selection in the fully retrieved runs.** Only questions whose two hops were both retrieved are kept (34% for E5, 18% for BM25), which favours questions with lexically salient hops.
- **Self-generated paraphrases.** Paraphrases and partials are generated by the same model that is probed. Validation removes rewrites that drop the sub-answer or copy the original, but generated text may still be easier for the model to read than human-written text. The independently authored paragraphs avoid this but are few (64 decisive increments on source-unseen questions).
- **One model.** All ESTB runs reported here use Qwen2.5-7B-Instruct.

Q Compute and reproducibility

Collection for one 2-hop model ($\approx 35k$ conditions) takes about 22 hours on 64 CPU cores (bf16, ≈ 300 prompt tokens/s) or about one hour on an H200 shared with another job; the 3–4-hop run took 1.3 GPU-hours; all causal experiments and text baselines together used about 7 GPU-hours. Probing runs on CPU in one to two hours per model. Every stage is deterministic given the seed (question sampling, insertion positions, false-answer choice, splits) and resumable. Code, configurations and the trajectory construction are released with the paper.